

ARUN S

Full-Stack Developer | MCA Student | ML Enthusiast

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Portfolio • LinkedIn • GitHub

CAREER OBJECTIVE

To secure a challenging Software Developer / Full-Stack Engineer role in a growth-oriented IT company where I can apply my expertise in React.js, Next.js, Node.js, and Python-based Machine Learning to build impactful, scalable digital products — while continuously advancing toward a senior engineering position.

PROFESSIONAL SUMMARY

Full-Stack Developer and MCA student at Sathyabama Institute of Science and Technology with 3+ end-to-end web application projects and a Python Machine Learning internship. Delivered production-grade platforms using React.js, Next.js, Node.js, Express.js, MongoDB, and PostgreSQL — reducing development turnaround by building reusable component libraries and REST APIs. Achieved 81.1% in BCA. Passionate about writing clean, maintainable code and creating high-performance user experiences that directly drive business outcomes.

TECHNICAL SKILLS

Frontend	React.js (JSX, Hooks), Next.js, HTML5, CSS3, Tailwind CSS, Framer Motion, JavaScript (ES6+)
Backend	Node.js, Express.js, REST API Design, JWT Authentication, Middleware Architecture
Databases	MongoDB, MySQL, PostgreSQL, Mongoose ODM
ML / AI	Python, Scikit-learn, Pandas, NumPy — Model Training, Data Analysis, Performance Optimization
Programming	Python, JavaScript, Java, C
Tools & DevOps	Git, GitHub, Vercel, VS Code, Postman, npm / yarn

PROJECTS

LEVEL UP – AI-Powered Career & Recruitment Platform

Tech Stack: Next.js · Node.js · Express.js · MongoDB · JWT Auth

- Architected a gamified recruitment platform for a 5-member team, replacing resume-only hiring with dynamic skill-progression profiles, cutting recruiter screening time by an estimated 40%.
- Built role-based dashboards (candidate, recruiter, admin) with AI-enhanced profile analysis and animated portfolio showcases, supporting 3 distinct user roles across a single codebase.
- Engineered a skill leveling and ranking engine with 10+ competency categories, enabling recruiters to rank applicants by verified scores rather than self-reported experience.
- Reduced page load friction by implementing lazy loading and reusable component architecture across 15+ pages, improving UI consistency and development speed by ~30%.

FurnitureMart – Full-Stack E-Commerce Web Application

Tech Stack: Node.js · Express.js · MongoDB · HTML/CSS/JS · JWT Auth

- Developed a full-stack furniture e-commerce platform handling 100+ product SKUs with category filtering, cart management, and secure checkout — reducing manual order errors to zero.
- Implemented end-to-end authentication (register, login, JWT sessions) with role-based route protection, covering 2 user roles and 8+ secured API endpoints.
- Built an admin panel with full CRUD for products, users, and orders, cutting product update time by ~50% compared to manual database operations.
- Designed and optimised MongoDB schemas with indexing on product and order collections, achieving sub-100ms average query response times.

Smart To-Do App – Task Management Application

Tech Stack: HTML · CSS · JavaScript · Local Storage API

- Built a responsive task management app with 3-level priority filters and Local Storage persistence, retaining 100% of user data across sessions without a backend.
- Delivered a clean, mobile-responsive UI with sub-50ms interaction response times for task creation, editing, and deletion — scoring 95+ on Lighthouse performance audit.

RF Signal Prediction System – ML Project

Tech Stack: Python · Scikit-learn · Pandas · NumPy · Matplotlib

- Trained Linear Regression and Random Forest models on a 1,000+ record dataset to predict RF signal strength, with Random Forest achieving 89% R² score.
- Reduced model RMSE by 22% over baseline through feature engineering, outlier removal, and hyperparameter tuning via 5-fold cross-validation.

INTERNSHIP EXPERIENCE

Machine Learning with Python Intern

Retech Solutions Pvt. Ltd., Tambaram West, Chennai | June 2024

- Built and deployed 3+ supervised ML models (classification & regression) using Python and Scikit-learn, achieving up to 89% accuracy on held-out test sets.
- Reduced data preprocessing time by ~35% by automating cleaning pipelines (missing value imputation, encoding, normalisation) using Pandas and NumPy across 5,000+ row datasets.
- Improved model F1-score by 18% through iterative hyperparameter tuning and k-fold cross-validation, documented in structured performance reports presented to senior engineers.
- Visualised model outputs and feature importance using Matplotlib and Seaborn, enabling the team to communicate findings clearly to non-technical stakeholders.

EDUCATION

Master of Computer Applications (MCA)	2025 – Present
Sathyabama Institute of Science and Technology, Chennai	Pursuing
Bachelor of Computer Applications (BCA)	2022 – 2025
Guru Nanak College, Chennai	81.10%
HSC – Class XII	2021 – 2022
SSV HR Sec School, Mathapattanam	66.5%
SSLC – Class X	2019 – 2020
SSV HR Sec School, Mathapattanam	61.2%

CERTIFICATIONS

- MongoDB Certified Developer – Proficiency in NoSQL schema design, aggregation pipelines, indexing, and performance tuning.
- Cybersecurity Essentials – Cisco Networking Academy; foundational knowledge in network security, threat mitigation, cryptography, and cybersecurity best practices.

KEY STRENGTHS

- Problem-Solving & Analytical Thinking – Consistently debug complex issues across frontend, backend, and ML pipelines.
- Agile Self-Learner – Rapidly adopt new frameworks and technologies through project-based learning and documentation.
- Team Collaboration – Contributed as a team member on academic and internship projects, practicing code reviews and peer feedback.
- Attention to Detail – Emphasis on clean code, UI precision, and thorough testing before deployment.